10. STUDY OF RELATION BETWEEN LENGTH OF THE GIVEN WIRE AND TENSION FOR A CONSTANT FREQUENCY USING SONOMETER

AIM

To study the relationship between the length of a given wire and tension for constant frequency using a sonometer

APPARATUS REQUIRED

Sonometer, tuning fork of known frequency, meter scale, rubber pad, paper rider, hanger with half – kilogram masses, wooden bridges.

FORMULA

The frequency of the fundamental mode of vibration of a string is given by,

$$n = \frac{1}{2l}\sqrt{\frac{T}{m}}\ Hz$$

If n is a constant, for a given wire (m is constant)

$$\frac{\sqrt{T}}{l} \text{ is constant.}$$

where $n \leftarrow$ Frequency of the fundamental mode of vibration of a string (Hz)

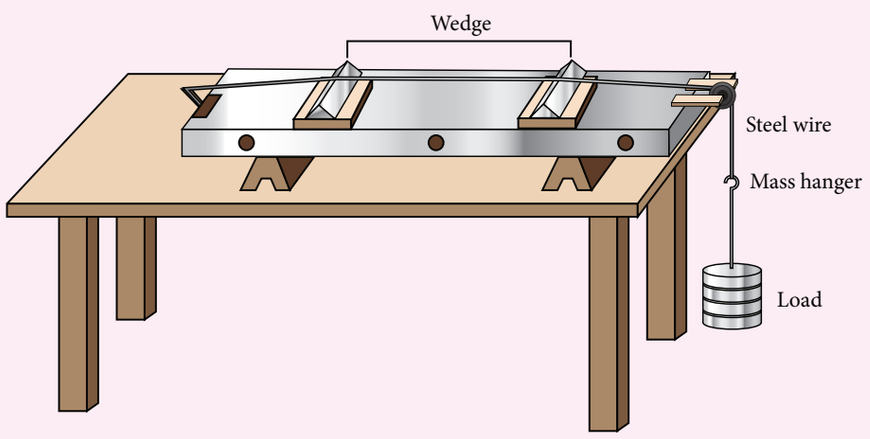
$m \leftarrow$ Mass per unit length of string ($kg\ m^{-1}$)

$T \leftarrow$ Tension in the string (including the mass of the hanger) = Mg (N)

$l \leftarrow$ Length of the string between the wedges (metre)

$M \leftarrow$ Mass suspended, including the mass of the hanger (kg)

DIAGRAM



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